

Velora — CPRI Hackathon Methodology Note

1. Objective

The pipeline audits noisy electrical test measurements relative to their operating context and cross-sensor consistency, predicts Validity_Label, and predicts Reference_Parameter for the hidden test set.

2. Data integrity and audit

The official participant PDF was programmatically parsed and compared against the repository CSVs by Test_ID. Training: 1,000/1,000 matched. Test: 350/350 matched. Cell-level mismatches: zero. Label mismatches: zero. Reference_Parameter mismatches: zero. Exact duplicate rows: zero; 12 observable duplicate groups involved 24 rows.

3. Missing values and anomalies

Training missing counts are S1=6, S2=2, S3=7, S4=29; test counts are S1=1, S2=3, S3=2, S4=11. S1–S3 are imputed using leakage-safe Ridge models. S4 is deliberately left missing and represented with a missingness flag. Suspicious duplicate observations and sentinel/anomalous measurements are retained rather than silently deleted.

4. Operating partitions

KMeans candidates k=4, 5, and 6 were compared. k=4 was selected for balance and interpretability. Training cluster sizes are 252, 265, 240, and 243; test sizes are 99, 87, 83, and 81. Stability was assessed using ARI (mean 0.4937, SD 0.2299). These are empirical partitions, not proven physical regimes.

5. Cross-sensor consistency

The authoritative consistency model is Ridge(alpha=1.0). Its out-of-fold performance is MAE 0.494530, RMSE 1.672792, and R² 0.893817. Residual behavior is used as contextual anomaly evidence rather than as a standalone global-outlier rule.

6. Validity classification

A frozen Random Forest is used for validity classification. Development/holdout separation is 800/200. The operational threshold is 0.236, selected from development folds without using the locked holdout. Primary locked-holdout confusion matrix: TN=170, FP=3, FN=3, TP=24. Invalid precision, recall and F1 are each 0.888889.

7. Reference Parameter prediction

The final regressor is GradientBoostingRegressor with n_estimators=100, max_depth=3, learning_rate=0.1, random_state=42. It is trained on the full training data only after model/validation decisions were frozen. Locked holdout metrics are MAE 1.257678, RMSE 3.613879, R² 0.887163, Median AE 0.435748, bias +0.038177, and maximum absolute error 31.016624.

8. Attention and audit scoring

The frozen attention specification is: 3×negative flag + 3×sentinel flag + 2×inconsistency flag + min(|z_S3|,10) + imputed count + missing count + 2×(1–2|P_invalid–0.5|) + 2×(1–support). The score is an operational audit heuristic. It does not directly consume ground-truth labels/targets at scoring time, but it incorporates predictions from the supervised validity model.

9. Leakage controls

Learned preprocessing and modeling operations are fitted within the appropriate training folds. Threshold selection is leakage-safe and the 200-row locked holdout is excluded from operational threshold optimization. The final production artifacts were frozen before ground-truth verification.

10. Validation and robustness

The project includes automated unit, reconciliation, adversarial, and dataset-verification tests. The final verification run reports 291 passed and 0 failed. CV model disagreement is treated as a diagnostic rather than a formal prediction interval. Support is not claimed to be a universal OOD detector.

11. Reproducibility

Random seed: 42. The verified production run uses Python 3.14.4 and the frozen project environment. Final production artifact hashes were checked for immutability during CP12.1.

12. Final submission integrity

Velora.csv contains 350 rows and exactly the required three columns: Test_ID, Validity_Label, Reference_Parameter. It contains no nulls or duplicate Test_IDs. SHA-256: 5586561015bec10b02ebf00a2491e42b9670ab6b8ce54aed89e0393e3dfaf67c.

13. Limitations

Operating clusters are empirical; attention is heuristic; support is not a universal OOD guarantee; CV disagreement is not a formal uncertainty interval; regression errors are asymmetric across validity groups; and the locked holdout is a finite sample.